

Approximating fixed size quantum correlations in polynomial time

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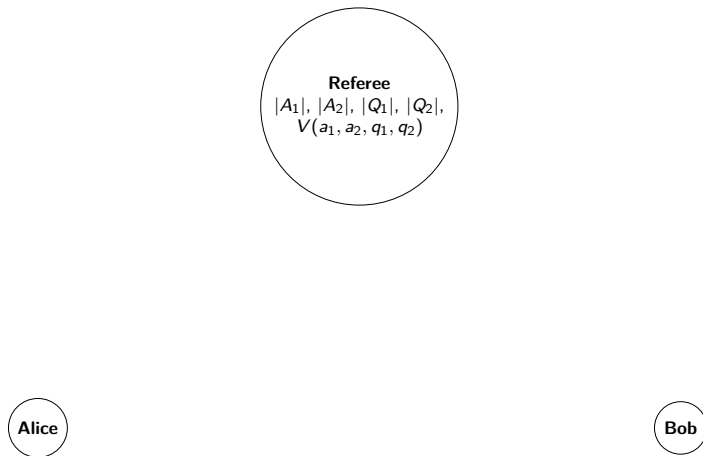
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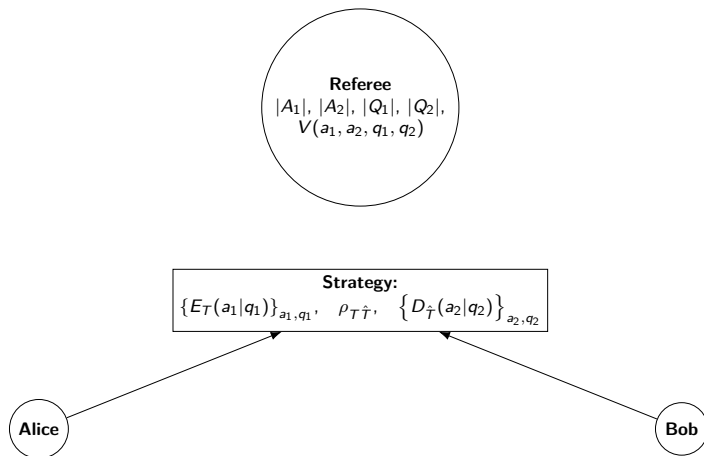
² *Univ Lyon, Inria, ENS Lyon, UCBL, LIP, Lyon, France*

05.03.2026, Quantum at the Dunes, Natal, Brazil

AQEC 2507.12326 , Non-local Games 2507.12302, GPTs 2601.15184

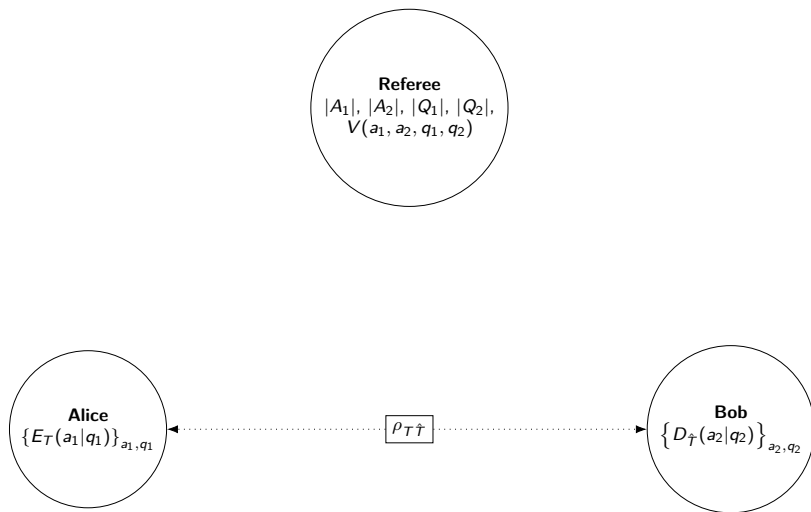
Non-local Games



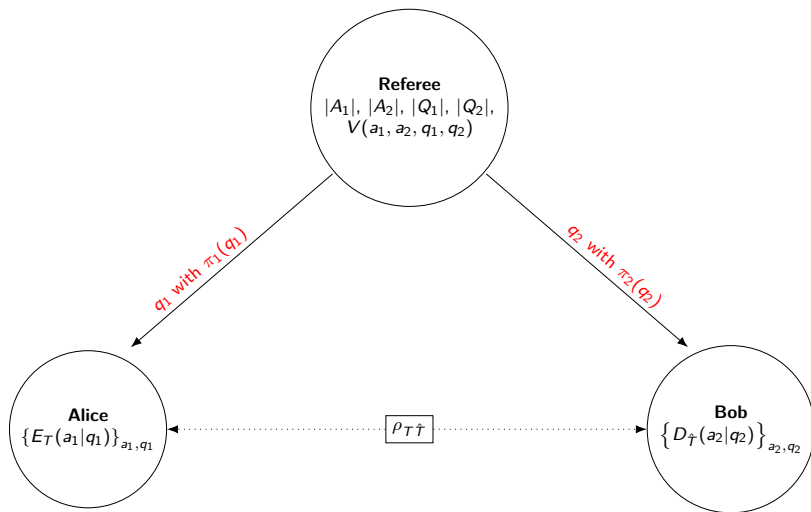


$T = 1$: Classical games [KRT08],
 $1 < T < \infty$: [NTV14; Nav+15; NV15; Jee+21]
 $T \rightarrow \infty$: unbounded games [BH17]

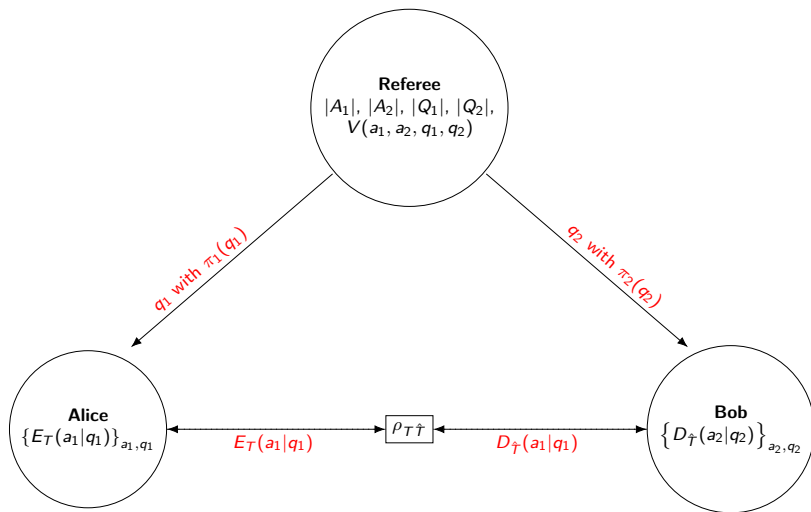
Free and fixed-sized non-local games



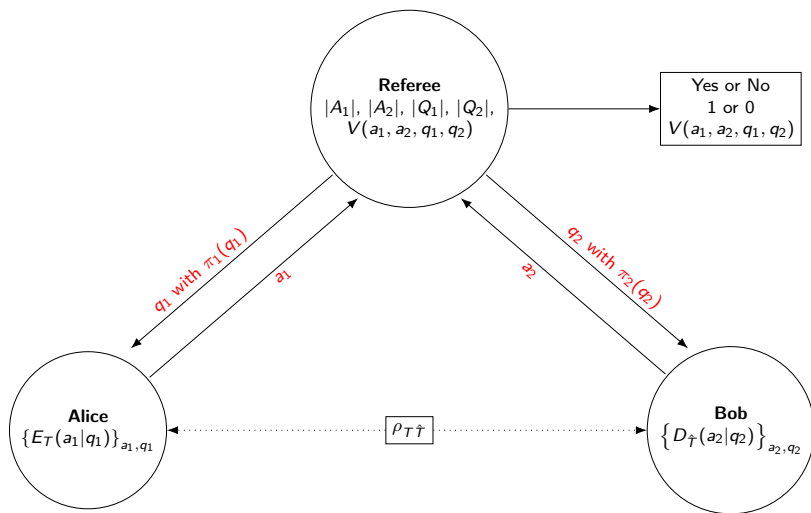
Free and fixed-sized non-local games



Free and fixed-sized non-local games



Free and fixed-sized non-local games



$$w_{Q(T)}(V, \pi) = |T| \max_{(\rho, E, D)} \text{Tr} \left[\left(V_{A_1 A_2 Q_1 Q_2} \otimes \rho_{T \hat{T}} \right) \left(E_{A_1 Q_1 T} \otimes D_{A_2 Q_2 \hat{T}} \right) \right]$$

$$\text{s.t.} \quad \rho_{T \hat{T}} \succcurlyeq 0, \quad \text{Tr} [\rho_{T \hat{T}}] = 1,$$

$$E_{A_1 Q_1 T} = \sum_{a_1, q_1} \pi_1(q_1) |a_1 q_1\rangle \langle a_1 q_1|_{A_1 Q_1} \otimes E_T(a_1 | q_1) \succcurlyeq 0,$$

$$E_{Q_1 T} = \sum_{q_1} \pi_1(q_1) |q_1\rangle \langle q_1|_{Q_1} \otimes \mathcal{I}_T,$$

$$D_{A_2 Q_2 \hat{T}} = \sum_{a_2, q_2} \pi_2(q_2) |a_2 q_2\rangle \langle a_2 q_2|_{A_2 Q_2} \otimes \frac{D_{\hat{T}}(a_2 | q_2)}{|\hat{T}|} \succcurlyeq 0,$$

$$D_{Q_2 \hat{T}} = \sum_{q_2} \pi_2(q_2) |q_1\rangle \langle q_1|_{Q_2} \otimes \frac{\mathcal{I}_{\hat{T}}}{|\hat{T}|},$$

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$$D_{A_2 Q_2 \hat{T}} = \sum_{a_2, q_2} \pi_2(q_2) |a_2 q_2\rangle \langle a_2 q_2|_{A_2 Q_2} \otimes \frac{D_{\hat{T}}(a_2|q_2)}{|\hat{T}|} \succcurlyeq 0,$$

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Problem

How well and at what computational cost can we approximate $w_{Q(T)}(V, \pi)$?

$$w_{Q(T)}(V, \pi) = |T| \max_{(\rho, E, D)} \text{Tr} \left[\left(V_{A_1 A_2 Q_1 Q_2} \otimes \rho_{T\hat{T}} \right) \left(E_{A_1 Q_1 T} \otimes D_{A_2 Q_2 \hat{T}} \right) \right]$$

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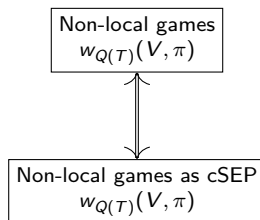
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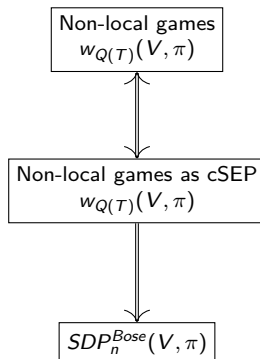
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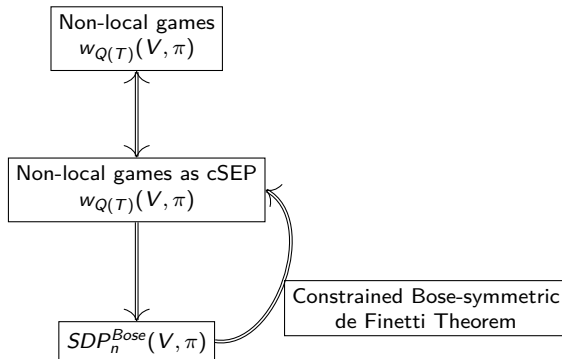
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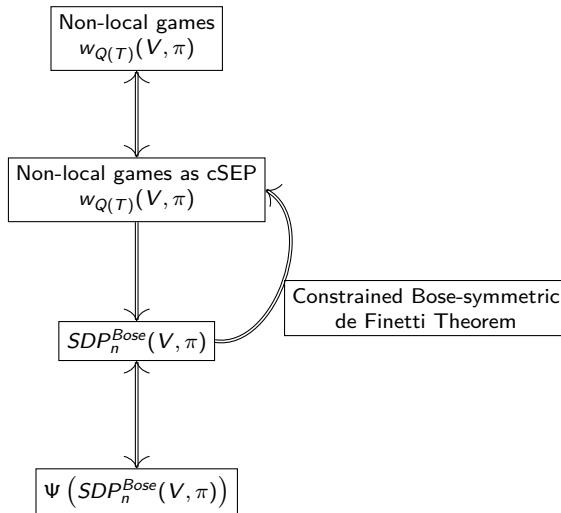
It is hard! C: PCP theorem [Aro+92; AS92], Q: [Cle+10; IKM09; Kem+11; BH17])

Non-local games
 $w_{Q(T)}(V, \pi)$

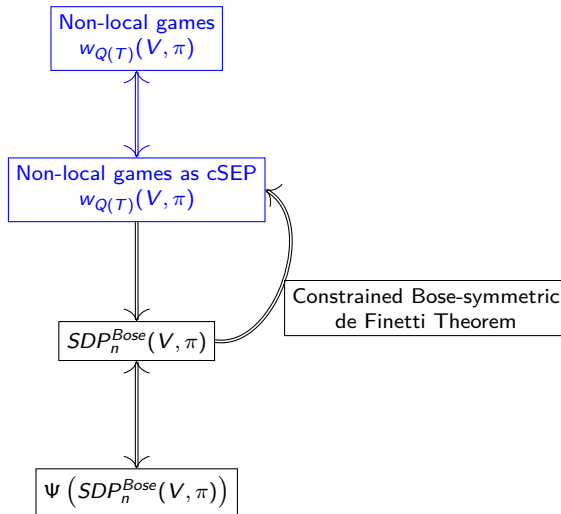








Non-local Games as constrained separability (cSEP) problems



$$w_{Q(T)}(V, \pi) = |T| \max_{(\alpha, D)} \text{Tr} \left[\left(V_{A_1 A_2 Q_1 Q_2} \otimes S_{\tilde{T} \hat{T}} \right) \left(\alpha_{A_1 Q_1} \tilde{T} \otimes D_{A_2 Q_2} \hat{T} \right) \right]$$

$$\text{s.t. } \alpha_{A_1 Q_1} \tilde{T} = \sum_{a_1, q_1} \pi_1(q_1) |a_1 q_1\rangle \langle a_1 q_1|_{A_1 Q_1} \otimes \alpha_{\tilde{T}}(a_1 | q_1) \succcurlyeq 0,$$

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- **Steering:** $\alpha_{\hat{T}}(a_1|q_1) := \text{Tr}_T \left[\left(\mathcal{I}_{\hat{T}} \otimes E_T(a_1|q_1) \right) \rho_{T \hat{T}} \right]$
- + some algebraic manipulations

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- Before: **Tripartite optimization** [Jee+21]

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- + some algebraic manipulations

- Before: **Tripartite optimization** [Jee+21]

- Now: Bipartite formulation yields **stronger approximation** results in terms of the dimension of quantum assistance

$$cSEP(G) := \max_{\rho_{AB} \in \mathcal{B}(AB)} \text{Tr}[G_{AB}(\rho_{AB})]$$

$$\begin{aligned} \text{s.t.} \quad \rho_{AB} &= \sum_{x \in \mathcal{X}} p(x) \rho_A^x \otimes \rho_B^x & \text{s.th. } \forall x \in \mathcal{X} : \\ \rho_A^x &\succeq 0, \rho_B^x \succeq 0, & \text{Tr}[\rho_A^x] = 1, \text{Tr}[\rho_B^x] = 1, \end{aligned}$$

Local constraints $\forall x \in \mathcal{X} :$

$$\Theta_{A_L \rightarrow C_{A_L}}(\rho_A^x) = W_{C_{A_L}} \otimes \rho_{A_R}^x, \quad \Upsilon_{B_L \rightarrow C_{B_L}}(\rho_B^x) = K_{C_{B_L}} \otimes \rho_{B_R}^x,$$

Fixed point constraints $\forall x \in \mathcal{X} :$

$$\Omega_{A \rightarrow A}(\rho_A^x) = \rho_A^x, \quad \Xi_{B \rightarrow B}(\rho_B^x) = \rho_B^x,$$

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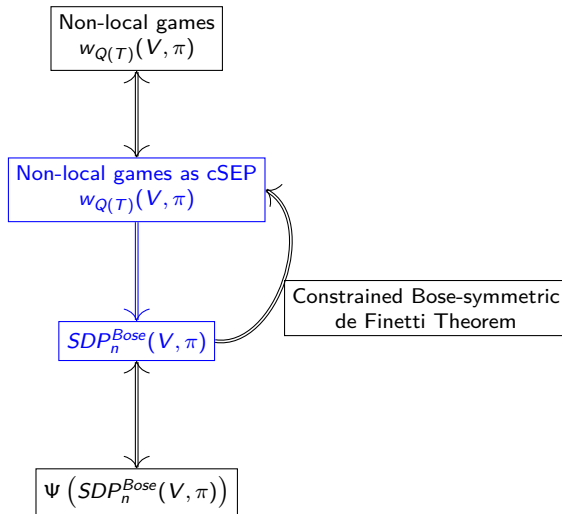
$$\Theta_{A_L \rightarrow C_{A_L}}(\rho_A^x) = W_{C_{A_L}} \otimes \rho_{A_R}^x, \quad \Upsilon_{B_L \rightarrow C_{B_L}}(\rho_B^x) = K_{C_{B_L}} \otimes \rho_{B_R}^x,$$

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- **Before:** No fixed point constraints, A_R and B_R one dimensional [Ber+21; Jee+21]
- SEP problems are already hard and non-linear [Ioa07]

SDP Approximation Hierarchy



Definition (Bose-symmetry)

A state $\rho_{B_1^n}$ is Bose-symmetric, if

$$U_{B_1^n}(\pi)\rho_{B_1^n} = \rho_{B_1^n}, \forall \pi \in \mathcal{S}_n. \quad (1)$$

Moreover, the state $\rho_{AB_1^n}$ is Bose-symmetric w.r.t. A , if

$$(\mathcal{I}_A \otimes U_{B_1^n}(\pi)) \rho_{AB_1^n} = \rho_{AB_1^n}, \forall \pi \in \mathcal{S}_n. \quad (2)$$

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- No difference for pure states (rank one)!
- $\vee^n(\mathcal{H}) := \{|\psi\rangle \in \mathcal{H}^{\otimes n} : U_{\mathcal{H}_1^n}(\pi) |\psi\rangle = |\psi\rangle, \forall \pi \in \mathcal{S}_n\}$
- One-and-a-half de Finetti [Chr+07], but **no constraints!**
- $\text{End}_{\mathbb{C}[\mathcal{S}_n]}(\mathcal{H}^{\otimes n}) \simeq \vee^n(\text{End}(\mathcal{H}^{\otimes n}))$ vs. $\text{End}_{\mathbb{C}[\mathcal{S}_n]}(\vee^n(\mathcal{H})) \simeq \text{End}(\vee^n(\mathcal{H}))$

Lemma

Consider $cSEP(G)$. Then,

$$\begin{aligned}
 SDP_n^{Bose}(G) &= \max_{\rho_{A(B\bar{B})_1^n}} \text{Tr}[G_{AB} \rho_{AB}] \\
 \text{s.t. } \rho_{A(B\bar{B})_1^n} &\succeq 0, \quad \text{Tr}[\rho_{A(B\bar{B})_1^n}] = 1, \quad \rho_{A(B\bar{B})_1^n} \text{ is Bose-symmetric w.r.t. } A, \\
 \Theta_{A_L \rightarrow C_{A_L}} \left(\rho_{A(B\bar{B})_1^n} \right) &= W_{C_{A_L}} \otimes \rho_{A_R(B\bar{B})_1^n}, \\
 \Omega_{A \rightarrow A} \left(\rho_{A(B\bar{B})_1^n} \right) &= \rho_{A(B\bar{B})_1^n}, \\
 \Upsilon_{(B_L)_n \rightarrow C_{B_L}} \circ \text{Tr}_{\bar{B}_n} \left(\rho_{(B\bar{B})_1^n} \right) &= K_{C_{B_L}} \otimes \rho_{(B\bar{B})_1^{n-1}(B_R)_n}, \\
 \Xi_{B_n \rightarrow B_n} \circ \text{Tr}_{\bar{B}_n} \left(\rho_{(B\bar{B})_1^n} \right) &= \rho_{(B\bar{B})_1^{n-1}B_n},
 \end{aligned} \tag{3}$$

such that

$$0 \leq SDP_n^{Bose}(G) - cSEP(G) \leq \min \{ f(A|B\bar{B}), f(B\bar{B}|\cdot) \} \sqrt{4 \ln(2)} \sqrt{\frac{\ln(d_A)}{n}}. \tag{4}$$

$$w_{Q(T)}(V, \pi) \Rightarrow \text{SDP}_n^{\text{Bose}}(T, V, \pi) \text{ (or } c\text{SEP}(G) \Rightarrow \text{SDP}_n^{\text{Bose}}(G))$$

- Separable states as convex mixture of pure product state? **No**, additional linear constraints...

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$$\rho_{A(B\bar{B})_1^n} := \sum_{x \in \mathcal{X}} p(x) \rho_A^x \otimes (|\psi^x\rangle \langle \psi^x|_{B\bar{B}})^{\otimes n}, \quad |\psi^x\rangle_{B\bar{B}} := (\sqrt{\rho_B^x} \otimes \mathcal{I}_{\bar{B}}) |\Psi\rangle_{B\bar{B}}$$

$$w_{Q(T)}(V, \pi) \Rightarrow SDP_n^{Bose}(T, V, \pi) \text{ (or } cSEP(G) \Rightarrow SDP_n^{Bose}(G))$$

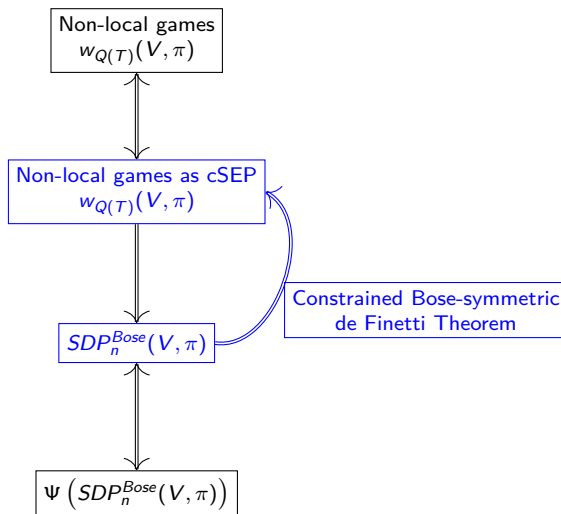
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- **Local purification scheme** $\rightarrow$ embedding of permutation-invariant states subject to linear constraints into the symmetric subspace of a higher-dimensional space, intersected with the corresponding constraints.

■

$$\rho_{A(B\bar{B})_1^n} := \sum_{x \in \mathcal{X}} p(x) \rho_A^x \otimes (|\psi^x\rangle \langle \psi^x|_{B\bar{B}})^{\otimes n}, \quad |\psi^x\rangle_{B\bar{B}} := (\sqrt{\rho_B^x} \otimes \mathcal{I}_{\bar{B}}) |\Psi\rangle_{B\bar{B}}$$

- Candidate is **Bose-symmetric** and **satisfies the linear constraints** in $SDP_n^{Bose}(T, V, \pi)$

De Finetti Theorems



Lemma: Constrained Bose-symmetric de Finetti Theorem

Let $\rho_{A(B\bar{B})_1^n}$ be Bose-symmetric with respect to A . Furthermore, consider linear mappings $\Theta_{A_L \rightarrow C_{A_L}}$, $\Omega_{A \rightarrow A}$, $\Upsilon_{B_L \rightarrow C_{B_L}}$, $\Xi_{B \rightarrow B}$ and operators $W_{C_{A_L}}$, $K_{C_{B_L}}$. Assuming

$$\begin{aligned} \Theta_{A_L \rightarrow C_{A_L}} \left(\rho_{A(B\bar{B})_1^n} \right) &= W_{C_{A_L}} \otimes \rho_{A_R(B\bar{B})_1^n}, & \Omega_{A \rightarrow A} \left(\rho_{A(B\bar{B})_1^n} \right) &= \rho_{A(B\bar{B})_1^n}, \\ \Upsilon_{(B_L)_n \rightarrow C_{B_L}} \circ \text{Tr}_{\bar{B}_n} \left(\rho_{(B\bar{B})_1^n} \right) &= K_{C_{B_L}} \otimes \rho_{(B\bar{B})_1^{n-1}(B_R)_n}, & \Xi_{B_n \rightarrow B_n} \circ \text{Tr}_{\bar{B}_n} \left(\rho_{(B\bar{B})_1^n} \right) &= \rho_{(B\bar{B})_1^{n-1}B_n}, \end{aligned}$$

then there exists a probability distribution $\{p(x)\}_{x \in \mathcal{X}}$ and states $\{\sigma_A^x\}_{x \in \mathcal{X}}$, $\{\sigma_{B\bar{B}}^x\}_{x \in \mathcal{X}}$ such that

$$\left\| \rho_{A(B\bar{B})} - \sum_{x \in \mathcal{X}} p(x) \sigma_A^x \otimes \sigma_{(B\bar{B})}^x \right\|_1 \leq \min \left\{ f(A|B\bar{B}), f(B\bar{B}|\cdot) \right\} \sqrt{4 \ln(2)} \sqrt{\frac{\ln(|\mathcal{A}|)}{n}}$$

where $f(A|B\bar{B}), f(B\bar{B}|\cdot)$ are minimal measurement distortions and for each $x \in \mathcal{X}$

$$\begin{aligned} \Theta_{A_L \rightarrow C_{A_L}}(\sigma_A^x) &= W_{C_{A_L}} \otimes \sigma_{A_R}^x, & \Omega_{A \rightarrow A}(\sigma_A^x) &= \sigma_A^x, \\ \Upsilon_{B_L \rightarrow C_{B_L}} \circ \text{Tr}_{\bar{B}} \left(\sigma_{(B\bar{B})}^x \right) &= K_{C_{B_L}} \otimes \sigma_{B_R}^x, & \Xi_{B \rightarrow B_n} \circ \text{Tr}_{\bar{B}} \left(\sigma_{(B\bar{B})}^x \right) &= \sigma_B^x. \end{aligned}$$

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- **ALSO:** New constrained de Finetti Theorem for **symmetric states** w/ application to new cSEP

- Conditioning with classical-quantum (cq) states

$$\rho_{A|z} = \frac{\text{Tr}_Z \left[\left(\mathcal{I}_A \otimes M_{B \rightarrow Z}^z \right) \rho_{AB} \right]}{\text{Tr} \left[\left(\mathcal{I}_A \otimes M_B^z \right) \rho_{AB} \right]} \quad (5)$$

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- The Self-decoupling Lemma [BH16, Lemma 2.1]: an information theoretic proof technique for symmetric cq-states

$$\mathbb{E}_{z_1^m} \left\| \rho_{AZ_{m+1}|z_1^m} - \rho_{A|z_1^m} \otimes \rho_{Z_{m+1}|z_1^m} \right\|_1 \leq \frac{2 \ln 2 \log |A|}{n}. \quad (6)$$

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- Replace classical-quantum states and use convexity

$$\left\| \rho_{A(B\bar{B})_{m+1}} - \sum_{z_1^m} p(z_1^m) \rho_{A|z_1^m} \otimes \rho_{(B\bar{B})_{m+1}|z_1^m} \right\|_1 \leq g(|A|, |B|) \sqrt{\frac{2 \ln 2 \log |A|}{n}}. \quad (7)$$

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- $g(|A|, |B|) = \min(f(A|B), f(B|)\cdot)$ can be bounded for informationally complete measurements
- Standard de Finetti representation theorems: No guarantee that the measure on states not satisfying the constraints is approximately zero

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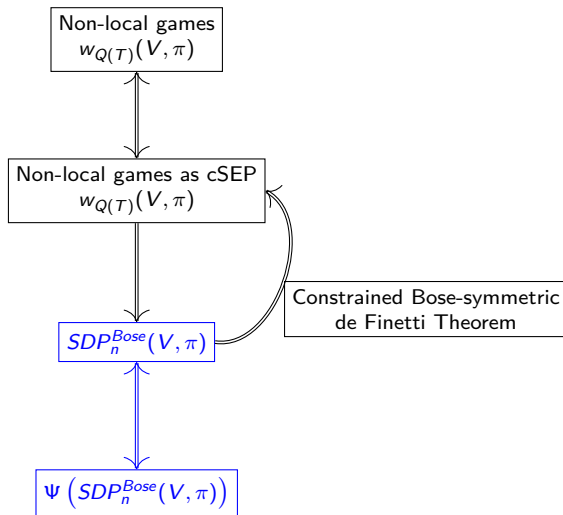
$$\begin{aligned}
 & \Upsilon_{(B_L)_{m+1} \rightarrow C_{B_L}} \circ \text{Tr}_{(\bar{B})_{m+1}} \left[\rho_{(B\bar{B})_{m+1}} |z_1^m \right] \\
 &= \frac{1}{N} \text{Tr}_{Z_1^m(B\bar{B})_{m+2}^n} \left[\left(M_{(B\bar{B})_1^m}^{z_1^m} \otimes \mathcal{I}_{C_{B_L}(B_R)_{m+1}(B\bar{B})_{m+2}^n} \right) \Upsilon_{(B_L)_{m+1} \rightarrow C_{B_L}} \circ \text{Tr}_{(\bar{B})_{m+1}} \left[\rho_{(B\bar{B})_1^n} \right] \right] \\
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$$\begin{aligned}
 & \Upsilon_{(B_L)_{m+1} \rightarrow C_{B_L}} \circ \text{Tr}_{(\bar{B})_{m+1}} \left[\rho_{(B\bar{B})_{m+1}|z_1^m} \right] \\
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 &= K_{C_{B_L}} \otimes \rho_{(B_R)_{m+1}|z_1^m}
 \end{aligned}$$

In paper: actually do the measurement to obtain inner points $\Rightarrow$ **Convergent inner sequence**

Efficient Approximation Algorithm



Restriction to polynomial in n operator space:

- $\Theta_{A_L \rightarrow C_{A_L}} \left(\rho_{A(B\bar{B})_1^n} \right), W_{C_{A_L}} \otimes \rho_{A_R(B\bar{B})_1^n} \in \text{End}_{\mathbb{C}[S_n]} \left(C_{A_L} \otimes A_R \otimes \vee^n (B\bar{B}) \right)$
- $\Omega_{A \rightarrow A} \left(\rho_{A(B\bar{B})_1^n} \right), \rho_{A(B\bar{B})_1^n} \in \text{End}_{\mathbb{C}[S_n]} \left(A \otimes \vee^n (B\bar{B}) \right)$
- $\Xi_{B_n \rightarrow B_n} \circ \text{Tr}_{\bar{B}_n} \left(\rho_{(B\bar{B})_1^n} \right), \rho_{(B\bar{B})_1^{n-1} B_n} \in \text{End}_{\mathbb{C}[S_n]} \left(\vee^{n-1} (B\bar{B}) \otimes B_n \right)$
- $\Upsilon_{(B_L)_n \rightarrow C_{B_L}} \circ \text{Tr}_{\bar{B}_n} \left(\rho_{(B\bar{B})_1^n} \right), K_{C_{B_L}} \otimes \rho_{(B\bar{B})_1^{n-1} (B_R)_n} \in \text{End}_{\mathbb{C}[S_{n-1}]} \left(C_{B_L} \otimes \vee^{n-1} (B\bar{B}) \otimes (B_R)_n \right)$

Problem

A priori,

$$\begin{aligned}
 SDP_n^{Bose}(G) &= \max_{\rho_{A(B\bar{B})_1^n} \in \text{End}_{\mathbb{C}[S_n]}(A \otimes \vee^n(B\bar{B}))} \text{Tr}[G_{AB} \rho_{AB}] \\
 \text{s.t. } \quad &\rho_{A(B\bar{B})_1^n} \succeq 0, \quad \text{Tr}[\rho_{A(B\bar{B})_1^n}] = 1, \\
 &\Theta_{A_L \rightarrow C_{A_L}}(\rho_{A(B\bar{B})_1^n}) = W_{C_{A_L}} \otimes \rho_{A_R(B\bar{B})_1^n}, \\
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 &\Xi_{B_n \rightarrow B_n} \circ \text{Tr}_{\bar{B}_n}(\rho_{(B\bar{B})_1^n}) = \rho_{(B\bar{B})_1^{n-1}B_n},
 \end{aligned}$$

is not given in a symmetry adapted basis. The objects in the canonical basis are of exponential size in n .

Question

Can we construct a symmetry-adapted SDP hierarchy, whose size scales polynomially with n , efficiently with respect to n ?

- $\text{End}_{\mathbb{C}[S_n]}(\vee^n(\mathcal{H})) \simeq \mathbb{C}^{m(n) \times m(n)}, \quad m(n) = \binom{n+|\mathcal{H}|-1}{n}$

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- A $*$ -isomorphism (change of basis) preserving semi-definiteness and the value of the SDP build from semistandard Young tableaux $\mathcal{T}_{\lambda,d}$:

$$\begin{aligned} \psi : \text{End}_{\mathbb{C}[S_n]}(\vee^n(\mathcal{H})) &\rightarrow \mathbb{C}^{m(n) \times m(n)} \\ A &\mapsto \left(\langle Au_\tau, u_\gamma \rangle_{S_n} \right)_{\tau, \gamma \in \mathcal{T}_{\lambda,d}} \end{aligned} \quad (8)$$

with

$$\mathbb{R}^{d_{\mathcal{H}}} \ni u_\tau = \sum_{\tau' \sim \tau} \bigotimes_{y \in Y((n))} e_{\tau'(y)} \quad (9)$$

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- Example:

$$\tau = \begin{array}{|c|c|c|} \hline 1 & 2 & 2 \\ \hline \end{array} \quad (10)$$

and corresponding Young symmetrizer

$$u_\tau = |011\rangle + |101\rangle + |110\rangle. \quad (11)$$

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Problem

Can we do this efficiently w.r.t. n ?

- Bose-symmetry $\Rightarrow$ "**Redundancies**" in canonical basis representation

-

$$\begin{bmatrix} b & & & \\ & a & a & \\ & a & a & \\ & & & c \end{bmatrix}$$

- **Efficient** representation of Bose-symmetric orbitals via contingency tables with fixed margins
- $\text{Poly}(n)$ -queries to an oracle

- New (Schur) basis: Bose-symmetric orbital averages

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Lemma

Let τ, γ be two semi-standard labellings of $Y((n))$. Given complex numbers $\{z_{\vec{t}, \vec{t}'}\}_{\vec{t}, \vec{t}' \in \mathcal{T}_{(n), d_{\mathcal{H}}}}$, then

$$\sum_{\vec{t}, \vec{t}' \in \mathcal{T}_{d_{\mathcal{H}}, n}} z_{\vec{t}, \vec{t}'} u_{\tau}^T C_{\vec{t}, \vec{t}'}^{\vee^n} u_{\gamma} \quad (12)$$

can be computed efficiently.

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Proof sketch: Use dual spaces [Pol20]:

$$\begin{aligned} \sum_{\vec{t}, \vec{t}' \in \mathcal{T}_{(n), d_{\mathcal{H}}}} u_{\tau}^T C_{\vec{t}, \vec{t}'}^{\vee n} u_{\gamma} \mu(\vec{t}, \vec{t}') &= \sum_{\vec{t}, \vec{t}' \in \mathcal{T}_{(n), d_{\mathcal{H}}}} g\left(C_{\vec{t}, \vec{t}'}^{\vee n}\right) \mu(\vec{t}, \vec{t}') \\ &= \sum_{\vec{t}, \vec{t}' \in \mathcal{T}_{(n), d_{\mathcal{H}}}} \binom{n}{\vec{t}} \cdot \binom{n}{\vec{t}'} \delta_{\vec{t}, \tau} \delta_{\vec{t}', \gamma} \mu(\vec{t}, \vec{t}') = \binom{n}{\tau} \cdot \binom{n}{\gamma} \mu(\tau, \mu) \end{aligned}$$

- Never construct exponentially sized $SDP_n^{Bose}(V, \pi)$
- $\Psi(SDP_n^{Bose}(V, \pi))$ can be efficiently constructed by:
 - All coefficients of the expressions in $SDP_n^{Bose}(V, \pi)$ are efficiently computable
 - Previous Lemma yields the efficient construction of $\Psi(SDP_n^{Bose}(V, \pi))$
- Since $\Psi(SDP_n^{Bose}(V, \pi)) = SDP_n^{Bose}(V, \pi)$, Bose-symmetric de Finetti bound translates to $\Psi(SDP_n^{Bose}(V, \pi))$

Theorem

For a two-player free non-local game with $|A|$ answers, $|Q|$ questions and quantum assistance of size $|T|$, there exists a $\text{poly}(\epsilon^{-1})$ costly transformation to an SDP with

$$\left(\epsilon^{-4} \cdot \text{poly}(|A|, |Q|, |T|)\right)^{(|A||Q||T|)^2},$$

many variables, which approximates the value of the game up to an additive ϵ -error.

Problem (Open)

Other uses of new cSEP framework?

Problem (Open)

It remains an open question whether other forms of Schur–Weyl duality admit analogous formulations with similar computational or structural utility.

Thank you for Listening!

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$$SDP_n^{Bose}(T, V, \pi) = |T| \max_{\rho} \text{Tr} \left[\left(V_{A_1 A_2 Q_1 Q_2} \otimes S_{T\bar{T}} \right) \rho_{(A_1 Q_1 T)(A_2 Q_2 \bar{T})} \right]$$

$$\text{s.t. } \rho_{(A_1 Q_1 T)(A_2 Q_2 \bar{T})} = \text{Tr} \left(\left((A_2 Q_2 \bar{T}) \left(\overline{(A_2 Q_2 \bar{T})} \right) \right)_2^n \left[\rho_{(A_1 Q_1 T)(A_2 Q_2 \bar{T})} \right]_1^n \right)$$

$$\rho_{(A_1 Q_1 T)(A_2 Q_2 \bar{T})} \left(\overline{(A_2 Q_2 \bar{T})} \right)_1^n \text{ is Bose-symmetric state w.r.t. } (A_1 Q_1 T)$$

$$\text{Tr}_{A_1} \left[\rho_{(A_1 Q_1 T)(A_2 Q_2 \bar{T})} \left(\overline{(A_2 Q_2 \bar{T})} \right)_1^n \right] = \sum_{q_1 \in Q_1} \pi_1(q_1) |q_1\rangle \langle q_1|_{Q_1} \otimes \rho_T \left((A_2 Q_2 \bar{T}) \left(\overline{(A_2 Q_2 \bar{T})} \right)_1^n \right)$$

$$\begin{aligned} & \text{Tr}_{(A_2)_n} \left(\overline{(A_2 Q_2 \bar{T})}_n \left[\rho_{(A_2 Q_2 \bar{T}) \left(\overline{(A_2 Q_2 \bar{T})} \right)_1^n} \right] \right) \\ &= \rho_{(A_2 Q_2 \bar{T}) \left(\overline{(A_2 Q_2 \bar{T})} \right)_1^{n-1}} \otimes \left(\sum_{q_2 \in Q_2} \pi_2(q_2) |q_2\rangle \langle q_2|_{Q_2} \otimes \frac{\mathcal{I}_{\bar{T}}}{|T|} \right). \end{aligned}$$